

Packet Doppler is used to measure and monitor the *delay distributions* of origin-destination (OD) pairs in an ISP network. A node could be an end-host, link, router or point of presence (PoP).

For each OD pair, tiny digests of real traffic (at both origin and destination) are listened to and recorded. The digests encode the time series information of the packets passing by.

Next, to extract all of the useful information out of packet digests, two decoding techniques are applied:

1. Sequential hypothesis testing (SHT) with Viterbi decoding
2. Sequential hypothesis testing (SHT) with iterative proportional fitting (IPF)

Keeping the structural constraints in the SLA measurements in mind, delay distributions are constructed, which are quite accurate.

The key advantages of Packet Doppler are:

- Passive technique
- Does not perturb real traffic
- Low storage and bandwidth overhead
- Detects burst events
- Detects packet loss events

Gossip protocol based monitoring

Gossip or epidemic protocols have emerged as a useful model to implement highly scalable, and resilient and reliable broadcast primitives. For reasons of scalability, participants in a gossip protocol maintain only a partial view of the system, from which they select peers to perform gossip exchanges.

The monitoring algorithm works by creating a network of nodes. Each server that is monitored represents a node. It also participates in monitoring the other nodes, by gossiping with them. Each node will periodically gossip its knowledge about the other nodes with a random node in the network.

The key concepts used by the algorithm are:

- **Heartbeat value:** Indicates the number of gossip rounds passed since the last communication with that node
- **Gossip list:** Contains a heartbeat value for each node
- **Gossip interval:** Time between two gossip messages sent by a node
- **Cleanup time:** Threshold value of the heartbeat before a node will be suspected to have failed
- **Suspect list:** List of suspects for each node, if the node is suspected to have failed
- **Suspect matrix:** 2D matrix, where each row is the suspect list of a node

When a node receives a message, it will update its own gossip list and suspect matrix if the received data is recent.

The protocol is generic enough and is used in NoSQL database design, in Akka router and cluster formations, among others, not to speak of distributed monitoring infrastructure.

Alerting

When a node detects that consensus is reached about a failure, it broadcasts this across the network. This can

happen multiple times for a single failure, as the consensus can be detected by multiple nodes simultaneously.

In order to prevent multiple alerts, only the live node with the lowest identifier will send the consensus broadcast and alert. In case this node fails, the next live node will take this role. The other nodes will know an alert has been sent when they receive the consensus broadcast.

With this approach, the failure detection time can be significantly reduced as there is no single point of failure.

Since the monitoring software is running on each node instead of being an external entity, the resource usage of the server can be monitored as well. This can potentially provide more information in case of a failure, and by monitoring the system health, possible failure causes can be detected early, helping to prevent downtime.

Network tomography

Network tomography (NT) is the study of the internal characteristics of the Internet using information derived from end nodes. According to the type of data acquisition and performance parameters being tracked, network tomography can be classified as:

1. Link-level parameter estimation based on end-to-end, path-level traffic measurements
2. Sender-receiver path level traffic intensity estimation based on link-level traffic measurements

NT is a comprehensive model for building inferential network monitoring techniques that require minimal cooperation from network elements.

Network Coding (NC) was developed with the aim of improving the throughput of networks. NC packets introduce topology-dependent correlation, which can be used to improve the accuracy and reduction of complexity in choosing monitoring paths. The introduction of NC to NT extends the bottleneck discovery to P2P networks. This has implications on traffic routing and other network performance metrics.

Network monitoring capabilities

What started as a model of end-point monitoring has grown to span a spectrum of network leveraged monitoring and management capabilities, largely driven by the complexity of the entire network and infrastructure. The various aspects of monitoring categorised below are seen in metrics driven service level agreements (SLA).

Availability and uptime monitoring:

- Network device monitoring
- Cisco monitoring
- Device health monitoring
- Switch monitoring
- Network mapping
- Custom network maps
- WAN RTT monitoring
- VoIP monitoring